

SECTION-A

1. a) Null space: $T(x_1, x_2) = (x_1 + x_2, x_1) = (0, 0)$

$\Rightarrow x_1 + x_2 = 0, x_1 = 0 \Rightarrow x_1 = 0, x_2 = 0$

$\Rightarrow N(T) = \{(0, 0)\}$ ——— (0.5)

Range: $R(T) = \{(x, y) \in \mathbb{R}^2 : x = x_1 + x_2, y = x_1, \forall (x_1, x_2) \in \mathbb{R}^2\}$ ——— (0.5)

1. b) Let U & V are two vector spaces over same field F . A function $T: U \rightarrow V$ is called L.T. if

(i) $T(u+v) = T(u) + T(v), \forall u, v \in U$

(ii) $T(au) = aT(u),$ where $a \in F$ & $u \in U$. ——— (1)

1. c) T is singular if $T(u) = 0 \Rightarrow u \neq 0$

$\Rightarrow T(x, y, z) = (x-y, y-z) = (0, 0)$

$\Rightarrow x = y$ & $y = z \Rightarrow u = (x, x, x) \neq 0$

Hence T is singular. ——— (1)

1. d) $(1, k, 5) = a_1(1, 2, 0) + a_2(0, -1, 1)$

$\Rightarrow 1 = a_1, k = 2a_1 - a_2, 5 = a_2$ ——— (0.5)

$\Rightarrow \boxed{k = -3}$ ——— (0.5)

1. e) vectors are L.D. $\Rightarrow \begin{vmatrix} 1+a & 1-a \\ 1-a & 1+a \end{vmatrix} = 0$ ——— (0.5)

$\Rightarrow \boxed{a = 0}$ ——— (0.5)

SECTION-B

2. a) Let $u = (x_1, y_1) \in \mathbb{R}^2$ s.t. $T(u) = (2x_1 - y_1, x_1 - y_1, -2x_1)$

$v = (x_2, y_2) \in \mathbb{R}^2$ s.t. $T(v) = (2x_2 - y_2, x_2 - y_2, -2x_2)$

Then, (i) $T(u+v) = T(x_1+x_2, y_1+y_2)$

$= (2x_1 + 2x_2 - y_1 - y_2, x_1 + x_2 - y_1 - y_2, -2x_1 - 2x_2)$

$= (2x_1 - y_1, x_1 - y_1, -2x_1) + (2x_2 - y_2, x_2 - y_2, -2x_2)$

$= T(u) + T(v)$ ——— (1)

(ii) $T(au) = T(ax_1, ay_1) = (2ax_1 - ay_1, ax_1 - ay_1, -2ax_1)$

$= a(2x_1 - y_1, x_1 - y_1, -2x_1)$

$= aT(u)$ ——— (1)

Hence T is a L.T.

b) T is invertible if it is one-one & onto or T is non-singular.

T is non-singular if $T(u) = 0 \Rightarrow u = 0$ ——— (0.5)

$$T(x, y, z) = (3x, x-y, 2x+y+z) = (0, 0, 0)$$

$$3x = 0 \Rightarrow \boxed{x=0}, \quad x-y=0 \Rightarrow \boxed{y=0}, \quad 2x+y+z=0 \Rightarrow \boxed{z=0}$$

$$\Rightarrow u = (x, y, z) = (0, 0, 0)$$

Hence T is non-singular $\Rightarrow T$ is invertible. ——— (0.5)

Find T^{-1}

$$\text{Let } T(x, y, z) = (r, s, t) \Rightarrow \boxed{(x, y, z) = T^{-1}(r, s, t)}$$

$$T(x, y, z) = (3x, x-y, 2x+y+z) = (r, s, t) \quad \text{——— (0.5)}$$

$$\Rightarrow 3x = r, \quad x-y = s, \quad 2x+y+z = t$$

$$\Rightarrow \boxed{x = \frac{r}{3}, \quad y = \frac{r}{3} - s, \quad z = s + t - r}$$

$$\Rightarrow T^{-1}(r, s, t) = \left(\frac{r}{3}, \frac{r}{3} - s, s + t - r \right) \quad \text{——— (0.5)}$$

2. c) $2 + x + x^2 = a_{11} + a_{12}(1+x) + a_{13}(1+x+x^2)$, on comparing coefficients

$$2 = a_{11} + a_{12} + a_{13}, \quad 1 = a_{12} + a_{13}, \quad 1 = a_{13} \Rightarrow \boxed{a_{13} = 1, a_{12} = 0, a_{11} = 1}$$

$$x + x^2 = a_{21} + a_{22}(1+x) + a_{23}(1+x+x^2)$$

$$\Rightarrow \boxed{a_{21} = -1, a_{22} = 0, a_{23} = 1} \quad \text{——— (0.5)}$$

$$x = a_{31} + a_{32}(1+x) + a_{33}(1+x+x^2)$$

$$\Rightarrow \boxed{a_{31} = -1, a_{32} = 1, a_{33} = 0} \quad \text{——— (0.5)}$$

Change of basis matrix P from B to B' is $\begin{bmatrix} 1 & -1 & -1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

(0.5)

SECTION-C

3. a) statement: Let U & V are vector spaces over same field and $T: U \rightarrow V$ is a L.T. from a finite dimensional space U to V , Then

$$\text{nullity of } T + \text{rank of } T = \dim(U). \quad \text{——— (0.5)}$$

Nullity $T(x, y, z) = (x+2y-z, y+z, x+y-2z) = (0, 0, 0)$

$$x+2y-z=0, \quad y+z=0, \quad x+y-2z=0$$

$$\Rightarrow x=3z, \quad y=-z \Rightarrow u = (3z, -z, z)$$

$$N(T) = \{(3, -1, 1)\}$$

$$\Rightarrow \boxed{\text{Nullity}(T) = 1} \quad \text{——— (1)}$$

is of Range :- Usual basis of $\mathbb{R}^3$ is $\{(1,0,0), (0,1,0), (0,0,1)\}$

$$T(1,0,0) = (1,0,1), \quad T(0,1,0) = (2,1,1), \quad T(0,0,1) = (-1,1,-2)$$

Image vectors spans T .

$$\therefore T = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ -1 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ Basis of Range is $\{(1,0,1), (0,1,-1)\}$

$$\Rightarrow \text{Rank}(T) = 2 \quad \text{--- (1)}$$

Also, $\dim(\mathbb{R}^3) = 3$

Hence Nullity $(T) + \text{Rank}(T) = 1 + 2 = 3 = \dim(\mathbb{R}^3)$ --- (0.5)

3. b.)

$$T(1,0,0) = (1,0) = a_{11}(1,2) + a_{12}(2,5)$$

$$1 = a_{11} + 2a_{12}, \quad 0 = 2a_{11} + 5a_{12} \Rightarrow \boxed{a_{11} = 5, a_{12} = -2}$$

$$T(0,1,0) = (1,1) = a_{21}(1,2) + a_{22}(2,5)$$

$$1 = a_{21} + 2a_{22}, \quad 1 = 2a_{21} + 5a_{22} \Rightarrow \boxed{a_{21} = 3, a_{22} = -1}$$

$$T(0,0,1) = (0,1) = a_{31}(1,2) + a_{32}(2,5)$$

$$0 = a_{31} + 2a_{32}, \quad 1 = 2a_{31} + 5a_{32} \Rightarrow \boxed{a_{31} = -2, a_{32} = 1}$$

$$\Rightarrow [T]_{B,B'} = \begin{bmatrix} 5 & 3 & -2 \\ -2 & -1 & 1 \end{bmatrix} \quad \text{--- (0.5)}$$

Let $v = (x, y, z) \in \mathbb{R}^3$, s.t. $(x, y, z) = a_1(1,0,0) + a_2(0,1,0) + a_3(0,0,1)$
 $\Rightarrow a_1 = x, a_2 = y, a_3 = z$

$$[v]_B = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{--- (0.5)}$$

$$T(v) = (x+y, y+z) = b_1(1,2) + b_2(2,5)$$

$$\Rightarrow x+y = b_1 + 2b_2, \quad y+z = 2b_1 + 5b_2$$

$$\Rightarrow b_1 = 5x + 3y - z, \quad b_2 = -2x - y + z$$

$$[T(v)]_{B'} = \begin{bmatrix} 5x + 3y - z \\ -2x - y + z \end{bmatrix} \quad \text{--- (0.5)}$$

$$[T]_{B,B'} [v]_B = \begin{bmatrix} 5 & 3 & -2 \\ -2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5x + 3y - z \\ -2x - y + z \end{bmatrix} = [T(v)]_{B'}$$

--- (0.5)

(i) Vectors of B are L.I., let $u_1 = (2, 1, 4)$, $u_2 = (1, -1, 2)$, $u_3 = (3, 1, -2)$
Let $a_1, a_2, a_3 \in \mathbb{R}$ are scalars s.t. $a_1 u_1 + a_2 u_2 + a_3 u_3 = 0$
 $a_1(2, 1, 4) + a_2(1, -1, 2) + a_3(3, 1, -2) = (0, 0, 0)$
 $2a_1 + a_2 + 3a_3 = 0$, $a_1 - a_2 + a_3 = 0$, $4a_1 + 2a_2 - 2a_3 = 0$
 $\Rightarrow a_1 = 0, a_2 = 0, a_3 = 0$

Vectors are L.I. if all scalars are zero.
and we get $a_1 = 0, a_2 = 0, a_3 = 0$ Hence u_1, u_2, u_3 are L.I.

(ii) Show that B spans vector space $\mathbb{R}^3(\mathbb{R})$
Let $v = (x, y, z) \in \mathbb{R}^3$ s.t. $v = a_1(2, 1, 4) + a_2(1, -1, 2) + a_3(3, 1, -2)$
 $x = 2a_1 + a_2 + 3a_3$, $y = a_1 - a_2 + a_3$, $z = 4a_1 + 2a_2 - 2a_3$

$$a_1 = \frac{y}{3} + \frac{z}{6}, \quad a_2 = \frac{x}{4} - \frac{2y}{3} + \frac{z}{24}, \quad a_3 = \frac{x}{4} - \frac{z}{8}$$

$$\Rightarrow (x, y, z) = \left(\frac{y}{3} + \frac{z}{6}\right)(2, 1, 4) + \left(\frac{x}{4} - \frac{2y}{3} + \frac{z}{24}\right)(1, -1, 2) + \left(\frac{x}{4} - \frac{z}{8}\right)(3, 1, -2)$$

Hence B spans V.

$\Rightarrow$ B is the basis of $\mathbb{R}^3(\mathbb{R})$

4. b) Basis of W :- Basis of W spanned by given vectors. Can be obtained by writing the vectors as rows of matrix and then reduce the matrix to row echelon form.

$$\text{Let } A = \begin{bmatrix} 1 & 4 & -1 & 3 \\ 0 & 2 & 1 & 5 \\ 2 & 1 & -3 & -1 \end{bmatrix} \quad R_2 - 2R_1$$

$$A \sim \begin{bmatrix} 1 & 4 & -1 & 3 \\ 0 & 2 & 1 & 5 \\ 0 & -7 & -1 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & -1 & 3 \\ 0 & 2 & 1 & 5 \\ 0 & 0 & 5 & 21 \end{bmatrix}$$

$$\therefore \text{Basis of } W = \{(1, 4, -1, 3), (0, 2, 1, 5), (0, 0, 5, 21)\}$$

$$\dim(W) = 3$$

Usual Basis of $\mathbb{R}^4 = \{(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1)\}$

Thus ~~to~~ extend the basis of W, we use usual basis.
Extended

$$\text{Basis of } \mathbb{R}^4 = \{(1, 4, -1, 3), (0, 2, 1, 5), (0, 0, 5, 21), (0, 0, 0, 1)\}$$

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